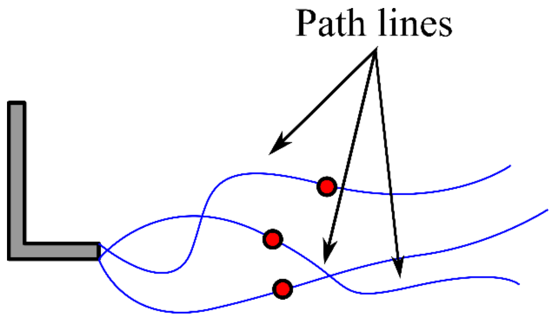
Pathline:

The pathline is the locus of fluid particles over the span of time. **A pathline is a line traced out in time by a given fluid particle as it moves through the flow field**. Two stream line cannot intersect each other or a streamline cannot intersect itself. Path lines can intersect themselves.

The actual path that a single fluid particle takes is referred to as the path line, i.e., it is the trajectory of a particular fluid particle. This is referred to as the **Lagrangian viewpoint** of the flow field. Experimentally, it can be achieved by tagging a fluid particle and tracing its motion throughout the flow field.

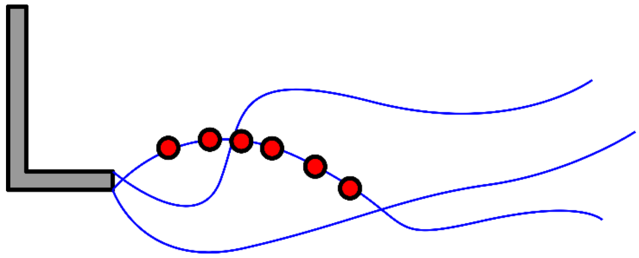


Streakline:

A streakline consists of all fluid particles that have previously passed through a specified point. This is essentially the same as injecting a dye or smoke continuously at a given location and observing how the dye or smoke moves along with the fluid motion. This is a technique that is often used in experiments to visualize the flow field.

The Streakline represents the position of all the fluid particles passed by a fixed point at a particular instant. It is specified by a fixed point in the flow field.

A line is created by the particles in a flow that have previously passed through the same point or a common point. For a steady flow path line is the same as the streak line and streamline.



Streamline:

Streamlines are useful as indicators of the instantaneous direction of fluid motion throughout the flow field. For example, regions of recirculating flow and separation of fluid off of a solid wall are easily identified by the streamline pattern. Streamlines cannot be directly observed experimentally except in steady flow fields, in which they are coincident with pathlines and streaklines.

- Streamline is the imaginary line drawn in the flow field such that the tangent on it at any point gives the direction of the velocity of that point at that instant.

- The velocity of each fluid particle flowing will remain constant with time in streamline flow.

- A streamline is defined as a line which is everywhere parallel to the local flow velocity vector (Velocity vector: $\vec{V} = \mathbf{u}\hat{\mathbf{i}} + \mathbf{v}\hat{\mathbf{j}} + \mathbf{w}\hat{\mathbf{k}}$)

- For a steady flow, the velocity field is dependent on **spatial coordinates** (x, y, z) only, $\vec{\mathbf{V}}(\mathbf{x}, \mathbf{y}, \mathbf{z})$

- For an unsteady flow, the velocity field is dependent on **spatial coordinates** (**x, y, z**) and **time** (**t**), $\vec{\mathbf{V}}(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{t})$

- Let an infinitesimal vector (infinitesimal arc length) be defined as the streamline as $\overrightarrow{\mathbf{ds}} = \mathbf{dx}\hat{\mathbf{i}} + \mathbf{dy}\hat{\mathbf{j}} + \mathbf{dz}\hat{\mathbf{k}}$

- Since this vector ($\overrightarrow{ds}$) is parallel to the velocity vector, it should satisfy the criterion $\overrightarrow{\mathbf{ds}} \times \vec{\mathbf{V}} = \mathbf{0}$

- Cartesian vector field (**2D flow**):

  Slope of $\overrightarrow{\mathbf{ds}} = \mathbf{dy}/\mathbf{dx}$ and slope of $\vec{\mathbf{V}} = \mathbf{v}/\mathbf{u}$

$$\overrightarrow{\mathbf{ds}} = \vec{\mathbf{V}} = \frac{\mathbf{dy}}{\mathbf{dx}} = \frac{\mathbf{v}}{\mathbf{u}}$$

  Rearranging the above equation, we get

$$\frac{\mathbf{dx}}{\mathbf{u}} = \frac{\mathbf{dy}}{\mathbf{v}} \rightarrow \textbf{Streamline equation}$$

  The above equation can be integrated to give the equation of a streamline once the velocity field (i.e., **u** and **v**) is known.

- Cartesian vector field (**3D flow**):

  $\overrightarrow{\mathbf{ds}} = \mathbf{dx}\hat{\mathbf{i}} + \mathbf{dy}\hat{\mathbf{j}} + \mathbf{dz}\hat{\mathbf{k}}$

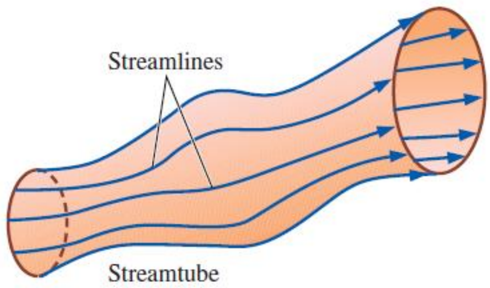
  $\vec{\mathbf{V}} = \mathbf{u}\hat{\mathbf{i}} + \mathbf{v}\hat{\mathbf{j}} + \mathbf{w}\hat{\mathbf{k}}$

$$\overrightarrow{\mathbf{ds}} \times \vec{\mathbf{V}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \mathbf{dx} & \mathbf{dy} & \mathbf{dz} \\ \mathbf{u} & \mathbf{v} & \mathbf{w} \end{vmatrix} = (\mathbf{wdy} - \mathbf{vdz})\hat{\mathbf{i}} + (\mathbf{udz} - \mathbf{wdx})\hat{\mathbf{j}} + (\mathbf{vdx} - \mathbf{udy})\hat{\mathbf{k}}$$

In general, a streamline, streak line and path line are not the same; however, they coincide when the flow is steady.

Streamtube:

A **streamtube** consists of a bundle of streamlines, much like a communications cable consists of a bundle of fiber-optic cables. Since streamlines are everywhere parallel to the local velocity, fluid cannot cross a streamline by definition.

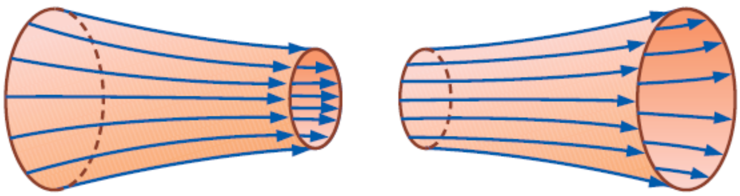


- By extension, *fluid within a streamtube must remain there and cannot cross the boundary of the streamtube*.

- Both **streamlines** and **streamtubes** are instantaneous quantities, defined at a particular instant in time according to the velocity field at that instant.

- At any instant in time, the mass flow rate passing through any cross-sectional slice of a given streamtube must remain the same.

- In a converging portion of an incompressible flow field, the diameter of the streamtube must decrease as the velocity increases in order to conserve mass and vice-versa.



(*a*) (*b*)

In an incompressible flow field, a streamtube (*a*) decreases in diameter as the flow accelerates or converges and (*b*) increases in diameter as the flow decelerates or diverges.

Stream Function:

It is defined as the scalar function of space and time, such that its partial derivative with respect to any direction gives the velocity component at right angles to that direction.

From the streamline equation:

$$\frac{\mathbf{dy}}{\mathbf{dx}} = \frac{\mathbf{v}}{\mathbf{u}}$$

Re-arranging the equation and integrate on both the sides, we get

$$\int \frac{dy}{v} = \int \frac{dx}{u}$$

$$g(x, y) + C_1 = h(x, y) + C_2$$

Let us assume that

$$\bar{\boldsymbol{\Psi}}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{g}(\mathbf{x}, \mathbf{y})}{\mathbf{h}(\mathbf{x}, \mathbf{y})} \text{ and } \mathbf{C} = \mathbf{C_2} - \mathbf{C_1}$$

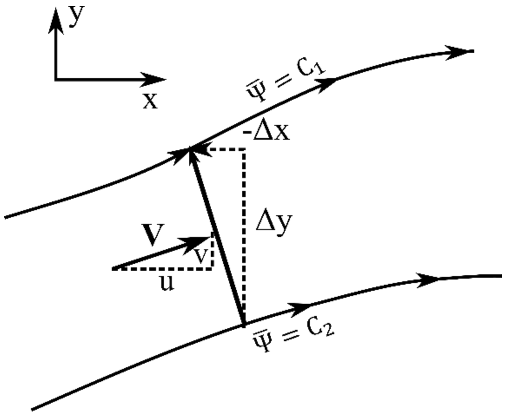
$$\bar{\Psi} - \text{represents a constructed stream function}$$

Stream Function is a **scalar function** used to define a **streamline**.

$$\bar{\boldsymbol{\Psi}}(\mathbf{x},\mathbf{y}) = \mathbf{C}\ (\text{In general on purpose})$$

$\bar{\boldsymbol{\Psi}}$ is **constant** along a **streamline.**

Two streamlines are separated by a distance $\boldsymbol{\Delta}\mathbf{n}$.



Physical Interpretation to the stream function $\bar{\boldsymbol{\Psi}}(\mathbf{x},\mathbf{y})$:

The difference between two streamlines is defined as,

$$\boldsymbol{\Delta}\bar{\boldsymbol{\Psi}} = \mathbf{C_2} - \mathbf{C_1} \equiv \dot{\mathbf{m}}\ (\text{mass flow }(\textbf{per unit depth perpendicular})\text{between streamlines})$$

Assume that these streamlines are close together (i.e., assume $\boldsymbol{\Delta}\mathbf{n}$ is small), such that the flow velocity ***V*** is a constant value across.

The mass flow through the stream tube per unit depth perpendicular to the page is

$$\boldsymbol{\Delta}\bar{\boldsymbol{\Psi}} \equiv \dot{\mathbf{m}} = \boldsymbol{\rho}\mathbf{V}\Delta\mathbf{n}(\mathbf{1})$$

$$\frac{\Delta\bar{\boldsymbol{\Psi}}}{\boldsymbol{\Delta}\mathbf{n}} = \boldsymbol{\rho}\mathbf{V}$$

Consider the limit of equation as $\boldsymbol{\Delta}\mathbf{n} \to \mathbf{0}$

$$\boldsymbol{\rho}\mathbf{V} = \lim_{\boldsymbol{\Delta}\mathbf{n}\to\mathbf{0}} \frac{\Delta\bar{\boldsymbol{\Psi}}}{\boldsymbol{\Delta}\mathbf{n}} = \frac{\partial\bar{\boldsymbol{\Psi}}}{\partial\mathbf{n}}$$

Notice that the directed normal distance $\boldsymbol{\Delta}\mathbf{n}$ is equivalent first to moving upward in the ***y*** direction by the amount $\boldsymbol{\Delta}\mathbf{y}$ and then to the left in the negative ***x*** direction by the amount $-\boldsymbol{\Delta}\mathbf{x}$**.**

$$\textbf{Mass flow }(\textbf{per unit depth}) = \Delta\bar{\boldsymbol{\Psi}} = \boldsymbol{\rho}\mathbf{V}\boldsymbol{\Delta}\mathbf{n} = \boldsymbol{\rho}\mathbf{u}\boldsymbol{\Delta}\mathbf{y}(\mathbf{1}) - \boldsymbol{\rho}\mathbf{v}\boldsymbol{\Delta}\mathbf{x}(\mathbf{1})$$

For **mass flow rate**, we multiply the **velocity** by its **orthogonal length**.

Assume $\boldsymbol{\Delta}\mathbf{n}$ is small, $\boldsymbol{\Delta} \to \mathbf{d}$

$$\mathbf{d}\bar{\boldsymbol{\Psi}} = \boldsymbol{\rho}\mathbf{u}\mathbf{d}\mathbf{y}(\mathbf{1}) - \boldsymbol{\rho}\mathbf{v}\mathbf{d}\mathbf{x}(\mathbf{1})$$

However, since $\bar{\boldsymbol{\Psi}} = \bar{\boldsymbol{\Psi}}(\boldsymbol{x},\boldsymbol{y})$, the chain rule of calculus states

$$\mathbf{d\bar{\Psi} = \frac{\partial\bar{\Psi}}{\partial x}dx + \frac{\partial\bar{\Psi}}{\partial y}dy}$$

Comparing the above equations, we get

$$\mathbf{\frac{\partial\bar{\Psi}}{\partial x} = -\rho v;\ \frac{\partial\bar{\Psi}}{\partial y} = \rho u}$$

If $\bar{\boldsymbol{\Psi}}(\boldsymbol{x}, \boldsymbol{y})$ is known for a given flow field, then at any point in the flow the products $\boldsymbol{\rho u}$ and $\boldsymbol{\rho v}$ can be obtained by differentiating $\bar{\boldsymbol{\Psi}}$ in the directions normal to ***u*** and ***v***, respectively.

In terms of polar coordinates,

$$\mathbf{\rho V_r = \frac{1}{r}\frac{\partial\bar{\Psi}}{\partial\theta};\ \rho V_\theta = -\frac{\partial\bar{\Psi}}{\partial r}}$$

In SI units, $\bar{\boldsymbol{\Psi}}$ is expressed as kilograms per second per meter or **kg/m.s** and $\Delta\bar{\boldsymbol{\Psi}}$ as cubic meters per second per meter or $\mathbf{m^2}/\boldsymbol{s}$**.**

For the **incompressible** flow**,** $\boldsymbol{\Psi} \equiv \bar{\boldsymbol{\Psi}}/\boldsymbol{\rho}$

$$\mathbf{\frac{\partial\Psi}{\partial x} = -v;\ \frac{\partial\Psi}{\partial y} = u}$$

$$\mathbf{V_r = \frac{1}{r}\frac{\partial\Psi}{\partial\theta};\ V_\theta = -\frac{\partial\Psi}{\partial r}}$$

The concept of the stream function is a powerful tool in aerodynamics, for two primary reasons. Assuming that $\bar{\boldsymbol{\Psi}}(\boldsymbol{x}, \boldsymbol{y})$ **(**or $\Psi(x, y)$) is known through the two-dimensional flow field, then:

1. $\bar{\boldsymbol{\Psi}}$ **= constant** gives the **equation of streamline.**
2. The **flow velocity** can be obtained by **differentiating** the **stream function** for incompressible and compressible flows.

Properties of stream function:

1. If stream function exists, it is a possible case of fluid flow which may be rotational or irrotational.
2. If the stream function satisfies Laplace equation, it is a case of irrotational flow.

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} = 0$$

References:

1. Anderson, J. (2011)*: Fundamentals of Aerodynamics (SI units)*. McGraw hill.
2. Cengel, Y., & Cimbala, J. (2013)*: Fluid mechanics fundamentals and applications (SI units)*. McGraw Hill.